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# AB Cloud Quantum Research: White Paper

One study: the algebraic AB (Aharonov–Bohm) architecture as a computing fabric that reads mathematics in real time and stores nothing

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# 1. Executive summary

AB Cloud Quantum Research is a single study built around one idea: the algebraic AB (Aharonov–Bohm) architecture as a computing fabric that reads mathematics — from operator spectra to zeta zeros — in real time and stores nothing. The study is organized in four levels (core → processor → verification → supplementary models), a shared formal layer (Lean 4 / Coq) and a theorem layer, every theorem carrying a certificate and a 2×2 four-dimension tile.

Flagship results: blind reading of all ζ zeros up to T = 1000 by two independent engines (649/649 Euler–Maclaurin + 639/639 Riemann–Siegel, cross-engine 0 bit-exact); exact compilation of unitary logic into AB clouds with 1−F ≤ 4.4e-16; the Guinand–Weil trace formula with the prime weight living in the heat trace of H_ξ (r = 0.9996); no-signaling nonlocality; and the formalized 'reads, does not carry' principle — a machine with no long-term memory.

*Table 1. Flagship numbers of the study.*

| Result | Numbers | Module |
|---|---|---|
| Blind ζ zeros (E17) | 649+639 blind; N(1000) = 649 | verifier_multilang |
| AB-cloud compiler (V20) | 1−F ≤ 4.4e-16 (4×4), 6.7e-16 (8×8) | processor_ab |
| Trace ↔ primes (V7/Th6) | r = 0.9996; control 0.76 | core_spectrum |
| Class B (E18) | ×59,000 in time vs 5 baselines | verifier_multilang |
| Verdicts | 18 PASS (pre-registered) | verdicts_master |

# 2. Architecture: the levels of one machine

The spectral core (modules/core_spectrum) supplies the mathematics: the operator H_ξ, campaigns V01–V19, theorems Th1–Th6, and the V13 obligation machine keeping the RH conclusion blocked until the analytic prerequisites are proved. The processor level (modules/processor_ab) executes it: the AB-cloud ISA, the closed-form Givens-QR compiler, and the 'reads, does not carry' principle. The verification level (modules/verifier_multilang) re-checks both layers with independent Julia engines: blind SHA-pinned protocols, cross-language replication, 18 pre-registered verdicts.

The formal layer pins the statements of all six theorems in Lean 4 and Coq (reductions A1–A6, no sorry/admit); the theorem layer makes them readable: 24 documents for Th1–Th6 and 12 for TS1–TS3, each with an embedded four-dimension tile. The supplementary study realtime_spinor_memory reduces the central claim to three explicit spinor phase models.

# 3. 'Reads, does not carry'

The key claim is that the AB-architecture computer needs no long-term storage of results: ζ zeros are reproduced from six constants of the trace formula at request time (12/12 exactly in the V23 preview; 649+639 zeros read blindly in E17), and the prime weight Λ(n)/√n provably lives in the heat trace of H_ξ. Storage = 0 is not an optimization but an architectural principle.

The strict formulation lives in the supplementary study: three spinor phase models ($\theta = \pi/2, \pi/3, \pi/7$) with a single volatile register. Theorem TS1 proves phase cycles close ($N = 2k$: 4, 6, 14), TS2 — that erasure is irreversible (reconstruction no better than the chance level $1/N$), TS3 — that the machine step costs $O(1)$ for all $t$ while an eternal log carries $\Omega(t)$ and misses deadlines. Hard real time and long-term memory are architecturally incompatible: volatility is the condition of determinism.

# 4. The honesty discipline

All thresholds are pre-registered and fixed before the run; verdicts compare mechanically. Detectors and validators are separated (blind protocols), detector outputs are SHA256-pinned before validation. Cross-language replication (Julia BigFloat ↔ Python mpmath, numpy ports of the compiler) reproduces the key numbers with 0.0–2.2e-16 disagreement. Negative results are published without retouching: lattice $\zeta$-correspondences are NOT found (V1/V3/V6/V16), vortices break Th5, the 2q KAK optimum is unreachable in the E18 ISA class. Errata are published in verdicts ($B/C_0$ of Th6; $\gamma_{12}$; ES2a/ES5a thresholds).

The conclusion about the Riemann hypothesis remains blocked by the V13 obligation machine: the program studies a mathematical model and claims no proof of RH. The Lean/Coq formalization guarantees this blocking rather than removing it.

# 5. Showcase and roadmap

The showcase: the flagship atlas (atlas/, a 2×2 tile), the timeline of 43 campaigns (timeline/), curated datasets (datasets/), a glossary (glossary/), and three web laboratories (the web_lab/ portal + 10 core tabs + 8 processor tabs). All laboratory computations run locally in the browser.

The roadmap is chained: processor P1 memory (V21) → P2 ALU (V22) → P3 reading (V23) → P4 nonlocality (V24) → P5 benchmarks → P6 Datasheet 1.0; verification O-J3 (QEC memory) → O-J4 (digital twin) → O-J5; core — the remainder of V19 stitching ($r_{align} \le 0.135$) and $\zeta$ monodromy; the spinor study — phases $\theta = \pi/k$ for $k > 7$ and task compilation into phase cycles.




*Fig. 1. The flagship atlas: blind ζ-zero reading, the exact compiler, trace ↔ primes, nonlocality (2×2 tile, 600 dpi, from committed verdicts).*